

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 06

Discrete Random Variable

Def: [Discrete RV] A random variable is said to have discrete distribution if there exists an atmost countable set $S_X \subset \mathbb{R}$ such that $P(X = x) > 0$ for all $x \in S_X$ and $\sum_{x \in S_X} P(X = x) = 1$. S_X is called the support of X .

Def: Define a function $f_X : \mathbb{R} \rightarrow [0, 1]$ by

$$f_X(x) = \begin{cases} P(X = x) & \text{if } x \in S_X \\ 0 & \text{otherwise.} \end{cases}$$

The function f_X is called the probability mass function of X .

Example 1, X is discrete, Example 2, X is discrete, Example 3, X_1 is not discrete, but X_2 is discrete.

Remarks

- For a discrete random variable X ,

$$F_X(x) = \sum_{\substack{y \in S_X \\ y \leq x}} f_X(y).$$

- For a discrete random variable X ,

$$f_X(x) = F_X(x) - F_X(x-).$$

Properties of PMF

$$\textcircled{1} \quad f_X(x) \geq 0 \text{ for all } x \in \mathbb{R}.$$

$$\textcircled{2} \quad \sum_{x \in S_X} f_X(x) = 1.$$

Theorem: Suppose a real valued function $h : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the following conditions:

$$\textcircled{1} \quad h(x) \geq 0 \text{ for all } x \in \mathbb{R}. \quad D = \{x : h(x) > 0\} \text{ is atmost countable.}$$

$$\textcircled{2} \quad \sum_{x \in D} h(x) = 1.$$

Then $h(\cdot)$ is a probability mass function of some discrete random variable.

Example

- ① (Bernoulli Distribution) $X \sim \text{Bernoulli}(p)$: $S_X = \{0, 1\}$,
 $f_X(0) = 1 - p$, $f_X(1) = p$.
- ② (Binomial Distribution) $X \sim \text{Bin}(n, p)$: $S_X = \{0, 1, \dots, n\}$,
 $f_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}$.
- ③ (Geometric Distribution) $X \sim \text{Geo}(p)$: $S_X = \{0, 1, \dots\}$,
 $f_X(k) = p(1 - p)^k$.
- ④ (Poisson Distribution) $X \sim \text{Poi}(\lambda)$ ($\lambda > 0$): $S_X = \{0, 1, \dots\}$,
 $f_X(k) = \frac{e^{-\lambda} \lambda^k}{k!}$.

Application

Example 1: Suppose that an airplane engine will fail, when in flight, with probability $1 - p$ independently from engine to engine. The airplane will make a successful flight if at least 50 percent of its engines remain operating. For what values of p is a four engine plane preferable to a two engine plane? (Ans: $p > 2/3$)

Continuous Random Variable

Def: A random variable is said to have a continuous distribution if there exists a non-negative integrable function $f_X : \mathbb{R} \rightarrow [0, \infty)$ such that

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

for all $x \in \mathbb{R}$. The function f_X is called the probability density function. The set $S_X = \{x \in \mathbb{R} : f_X(x) > 0\}$ is called support of X .

Example 2: (Exponential Distribution: $\text{Exp}(\lambda)$)

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Example 3: (Uniform Distribution: $U(a, b)$)

$$F_X(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x < b \\ 1 & \text{if } x \geq b. \end{cases}$$

Example 4: (Normal Distribution: $N(\mu, \sigma^2)$)

$$F_X(x) = \int_{-\infty}^x \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt \quad \text{if } -\infty < x < \infty.$$

Remarks on CRV

- ▶ For a continuous random variable X , $P(X = a) = 0$ for all $a \in \mathbb{R}$.
- ▶ CDF of a continuous random variable is continuous.
- ▶ PDF is not unique.
- ▶ Support of a continuous random variable is not unique.
- ▶ $P(a \leq X \leq b) = \int_a^b f_X(t) dt$.
- ▶ $f_X(x)$ is not $P(X = x)$.

Properties of PDF

$$\textcircled{1} \quad f_X(x) \geq 0 \text{ for all } x \in \mathbb{R}.$$

$$\textcircled{2} \quad \int_{-\infty}^{\infty} f_X(x) = 1.$$

Theorem: Suppose a real valued function $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the following conditions:

$$\textcircled{1} \quad g(x) \geq 0 \text{ for all } x \in \mathbb{R}.$$

$$\textcircled{2} \quad \int_{-\infty}^{\infty} g(x) dx = 1.$$

Then $g(\cdot)$ is a probability density function of some continuous random variable.

RV which is neither discrete nor continuous

Consider the random variable X whose distribution function is given by

$$F_X(x) = \begin{cases} 0 & \text{if } x < -1 \\ x + 1 & \text{if } -1 \leq x < -1/2 \\ 1 & \text{if } x \geq -1/2. \end{cases}$$

Observe that $F_X = 1/2F_1 + 1/2F_2$ where F_1 and F_2 are distribution functions given by

$$F_1(x) = \begin{cases} 0 & \text{if } x < -1 \\ 2(x + 1) & \text{if } -1 \leq x < -1/2 \\ 1 & \text{if } x \geq -1/2. \end{cases}$$

$$F_2(x) = \begin{cases} 0 & \text{if } x < -1/2 \\ 1 & \text{if } x \geq -1/2. \end{cases}$$